Meeting Notes

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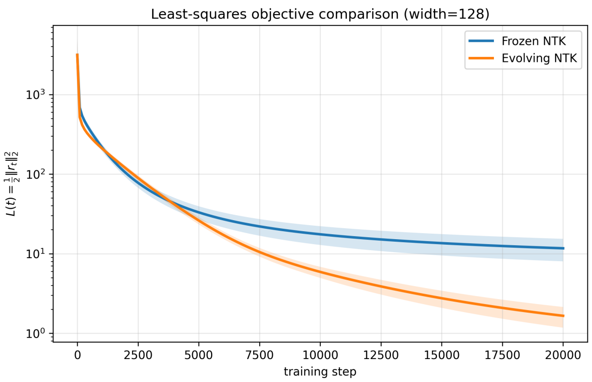
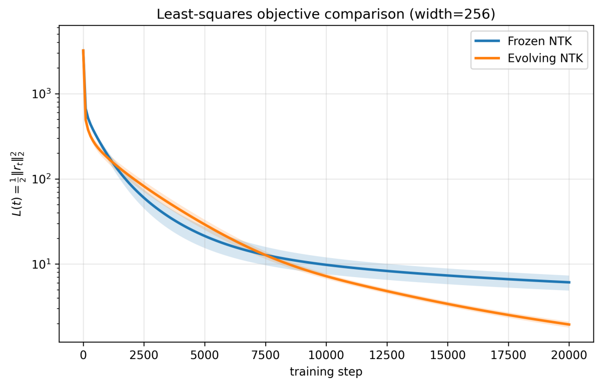
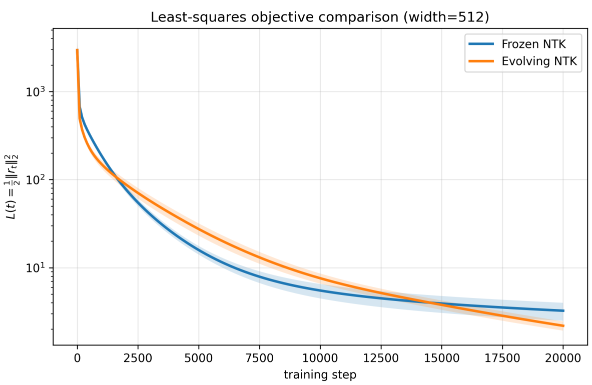
The question is why the evolving NTK helps at smaller widths, but not at larger widths.



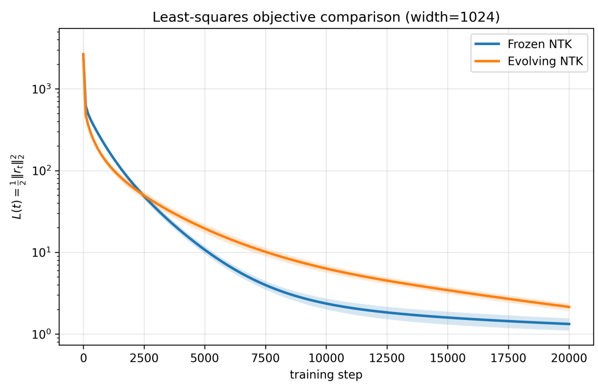
(a) Width 128.



(b) Width 256.



(c) Width 512.



(d) Width 1024.

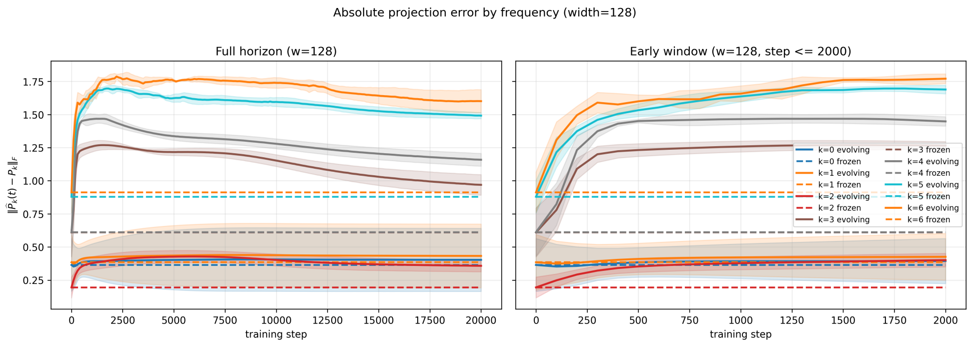
Figure 1: Least-squares objective comparison across widths. The evolving NTK gives a clear improvement at widths 128 and 256. The improvement is much smaller at width 512, and by width 1024 the frozen NTK gives the lower final objective.

The basic pattern is a width-dependent crossover. At small width, allowing the kernel to evolve improves optimization. At large width, the frozen kernel is already strong enough, and evolving the kernel no longer helps. To understand this, it is natural to first ask whether the evolving dynamics changes the Fourier subspaces themselves.

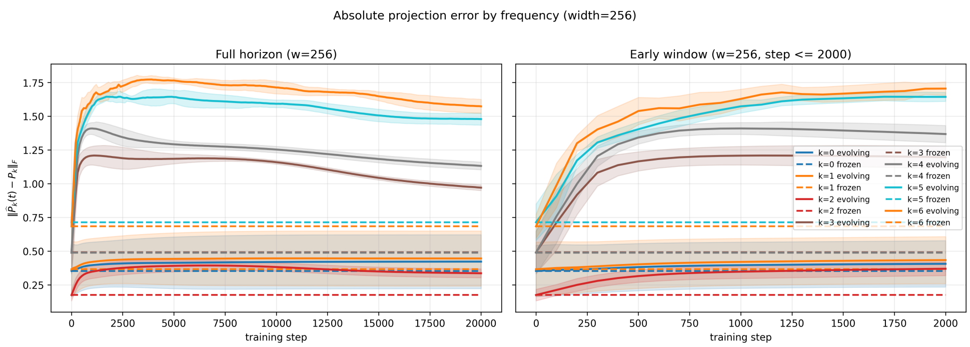
We therefore start with the projection error

$$\|\widehat{P}_k(t) - P_k\|_F,$$

which measures how far the learned projector for frequency k moves away from the reference Fourier projector.

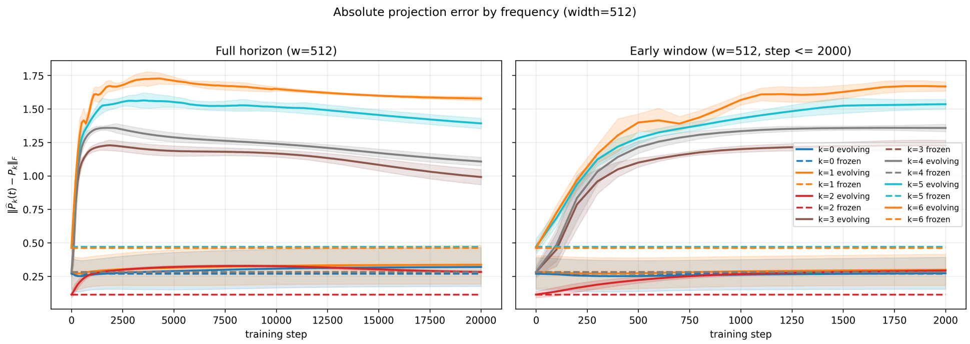


(a) Width 128.

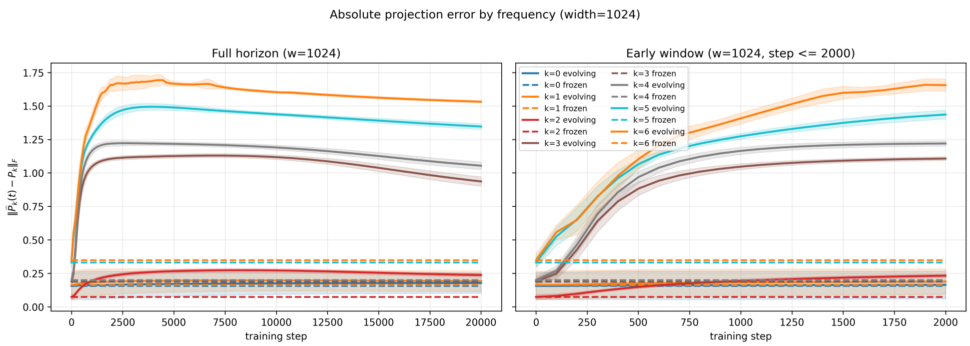


(b) Width 256.

Figure 2: Absolute projection error by Fourier subspace for widths 128 and 256.

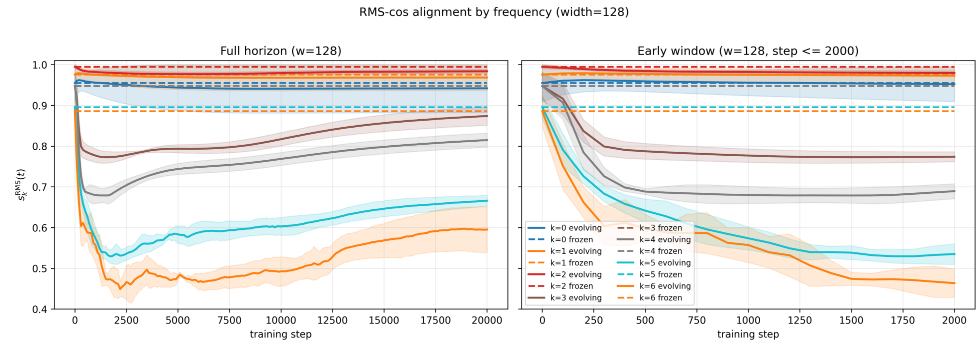


(a) Width 512.

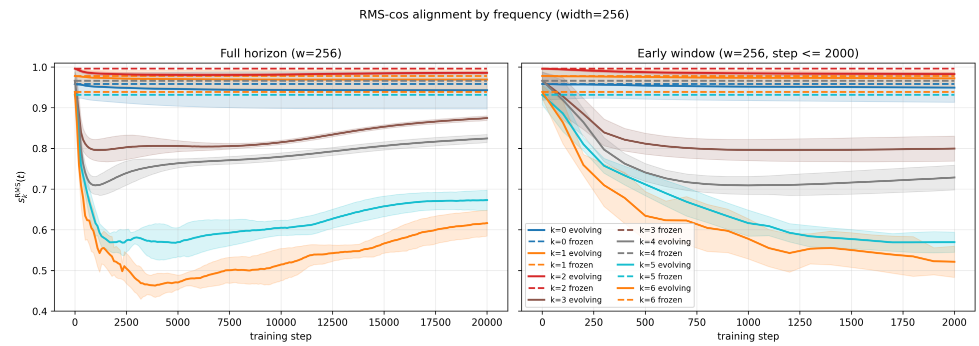


(b) Width 1024.

Figure 3: Absolute projection error by Fourier subspace for widths 512 and 1024.

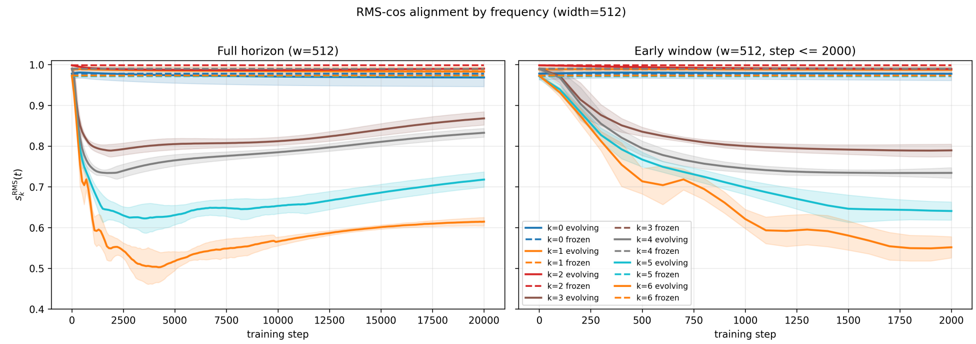


(a) Width 128.

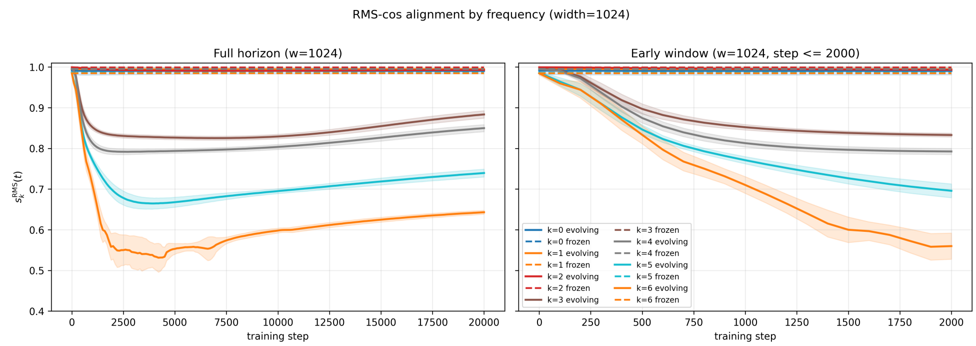


(b) Width 256.

Figure 4: RMS-cos alignment by Fourier subspace for widths 128 and 256.



(a) Width 512.



(b) Width 1024.

Figure 5: RMS-cos alignment by Fourier subspace for widths 512 and 1024.

The projection-error and cosine alignment plots show a consistent geometric trend. Under the evolving dynamics, several higher-frequency subspaces move away from their Fourier reference subspaces. The projection-error plots show this as an increase in $\|\widehat{P}_k(t) - P_k\|_F$, while the cos alignment plots show it as a drop in subspace alignment.

This rules out a simple explanation based on improved Fourier geometry. The evolving NTK does not help because it becomes more Fourier-aligned. If anything, it becomes less Fourier-aligned, especially for higher frequencies. The main difference across widths is the starting point. At low width, the frozen NTK already starts with worse Fourier geometry; at high width, it starts much closer to the Fourier reference. Thus, at high width, feature learning moves the kernel away from a better initial geometry.

Therefore, projection error and cosine alignment should be read mainly as geometry diagnostics. They show that feature learning changes the Fourier subspaces, but they do not by themselves explain why the evolving model reaches lower loss at small width. To understand the loss, we also need to know how strongly the kernel acts in the directions being fitted. This is the distinction between geometry and strength.

A small toy example makes this point clearer. Let

$$\mathcal{F}_1 = \operatorname{span}\{e_1, e_2\}$$

be a Fourier subspace, and let e_3 represent another frequency. Rotating the basis inside $\mathcal{F}_1$ does not change the subspace, and therefore does not change the projector P_1. Projection error appears only when the subspace itself tilts into another frequency direction. For example, suppose the learned subspace is

$$\widehat{\mathcal{F}}_1 = \operatorname{span}\{\cos\theta\, e_1 + \sin\theta\, e_3,\ e_2\}.$$

Then $\widehat{\mathcal{F}}_1 \neq \mathcal{F}_1$, and

$$\|\widehat{P}_1 - P_1\|_F = \sqrt{2}\sin\theta.$$

So this is genuinely worse Fourier alignment: the learned frequency block is no longer purely the original Fourier block.

However, worse alignment does not automatically mean slower loss decay. To see this, consider the two-dimensional slice $\mathrm{span}\{e_1, e_3\}$, where e_1 is a Fourier direction and e_3 represents a direction outside the original Fourier subspace. Suppose the residual is mainly in the Fourier direction

$$r = e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

A frozen, Fourier-aligned kernel may act on this slice as

$$K_{\text{frozen}} = \begin{pmatrix} 1 & 0 \\ 0 & 0.1 \end{pmatrix}.$$

Then

$$\langle e_1, K_{\text{frozen}} e_1 \rangle = 1.$$

Now suppose the evolving kernel tilts the leading eigendirection away from e_1 and toward e_3. Let

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The first tilted eigenvector is

$$u_1 = \cos\theta\, e_1 + \sin\theta\, e_3.$$

This is not a rotation inside the same Fourier subspace. It is a tilt into another frequency direction, so it would increase projection error.

Assume that the evolving kernel is diagonal in this tilted eigenbasis, with a larger leading eigenvalue:

$$K_{\text{evolving}} = R_\theta \begin{pmatrix} 3 & 0 \\ 0 & 0.1 \end{pmatrix} R_\theta^\top.$$

In the original (e_1, e_3) basis, this is

$$K_{\text{evolving}} = \begin{pmatrix} 3\cos^2\theta + 0.1\sin^2\theta & (3-0.1)\sin\theta\cos\theta \\ (3-0.1)\sin\theta\cos\theta & 3\sin^2\theta + 0.1\cos^2\theta \end{pmatrix}.$$

So for the same residual $r = e_1$,

$$\langle e_1, K_{\text{evolving}} e_1 \rangle = 3\cos^2\theta + 0.1\sin^2\theta.$$

For $\theta = 30^\circ$, this gives

$$3 \cdot 0.75 + 0.1 \cdot 0.25 = 2.275,$$

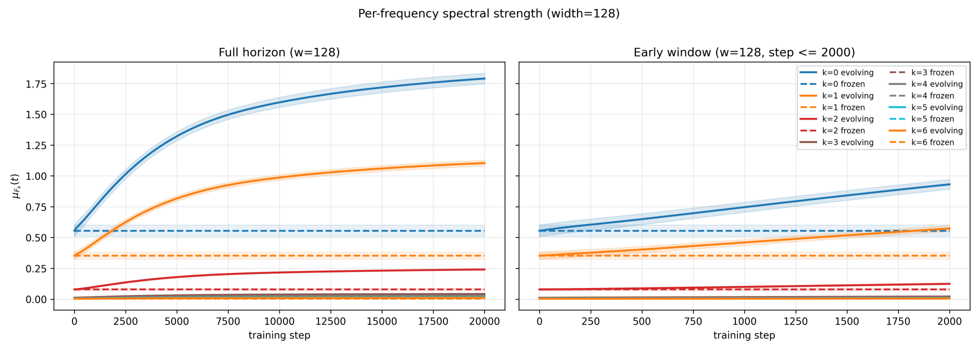
which is larger than the frozen value 1. Thus the evolving kernel can be less Fourier-aligned and still remove this residual direction faster, because the larger eigenvalue more than compensates for the tilt.

The example is not meant to prove that this is exactly what happens in the experiment. It only explains why the geometry plots are incomplete. Projection error and RMS-cos alignment measure whether the Fourier subspaces move. They do not measure whether the kernel becomes stronger along the directions that still matter for the residual. This motivates the next diagnostic: spectral strength.

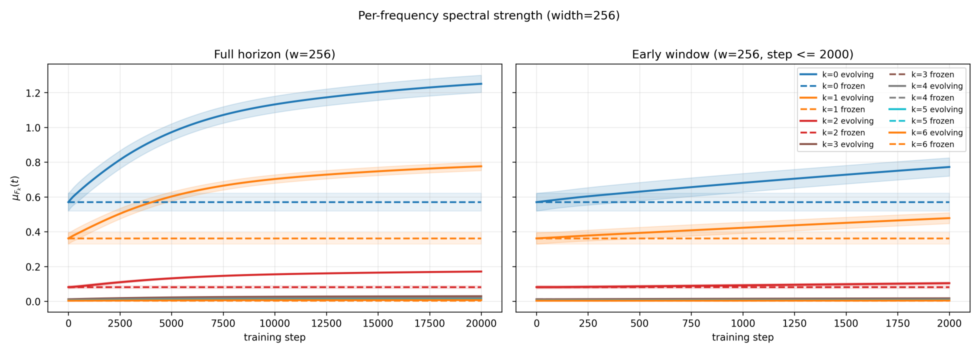
For each Fourier block $\mathcal{F}_k$, we summarize the average kernel strength by

$$\mu_{\mathcal{F}_k}(t) = \frac{1}{d_k}\,\mathrm{tr}(P_k K_t P_k)\,,$$

where P_k is the projector onto the k-th Fourier subspace and $d_k = \dim(\mathcal{F}_k)$. This is not an individual eigenvalue; it is the average spectral strength inside the subspace.

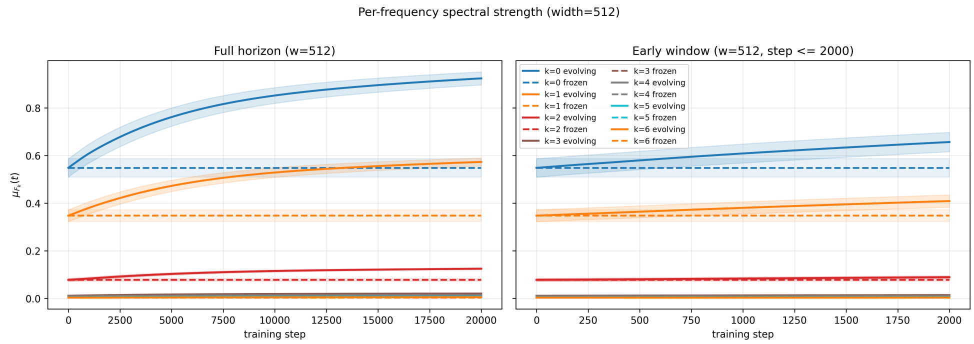


(a) Width 128.

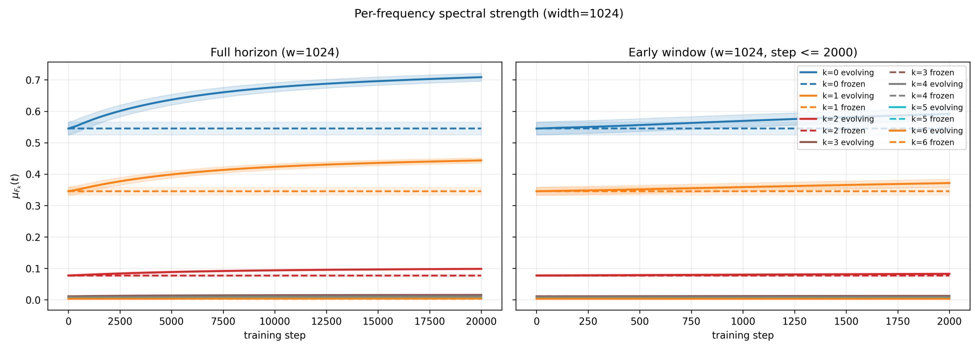


(b) Width 256.

Figure 6: Per-frequency spectral strength for widths 128 and 256.



(a) Width 512.



(b) Width 1024.

Figure 7: Per-frequency spectral strength for widths 512 and 1024.

The spectral-strength plots give the missing part of the story. The evolving NTK increases the average kernel strength in the low-frequency blocks, especially for $k = 0$ and $k = 1$, with a smaller effect for $k = 2$. This increase is largest at small width and becomes much weaker as width grows.

This suggests a simple tradeoff. The projection-error and alignment plots show that feature learning moves the Fourier geometry, which is a cost. The spectral-strength plots show the possible benefit: at low width, this movement comes with a large increase in low-frequency kernel strength. Since these low frequencies are important for the target, the evolving kernel can reduce the loss faster despite worse Fourier alignment.

At high width, the frozen NTK already starts from a better Fourier geometry, and the additional spectral-strength gain from feature learning is smaller. Then the benefit of evolving the kernel is no longer large enough to compensate for the geometric drift. This gives a plausible explanation for the crossover: feature learning helps at low width because spectral reweighting wins, but at high width the frozen NTK is already strong and stable enough that further evolution becomes less useful.

The next check should be residual-weighted spectral strength. The current plots measure average strength in each Fourier block, but they do not say how much residual is still present in that block. A more direct diagnostic would track both

$$\|P_k r_t\|^2 \qquad \text{and} \qquad \langle P_k r_t, K_t P_k r_t \rangle.$$

This would show whether the evolving kernel is becoming stronger exactly where the remaining residual is concentrated.